Lecture 1: OPERATORS IN MATHEMATICAL PHYSICS

1. Operator

$$\hat{A}: \ u(x) \longrightarrow v(x), \qquad \hat{A}u(x) = v(x)$$

An operator acts on a function $u(x)$ and produces a new function $v(x)$.

2. Linear Operator

An operator $\hat{L}$ is called *linear* if

$$\hat{L}(\alpha u + \beta v) = \alpha \hat{L} u + \beta \hat{L} v$$

Most operators in mathematical physics are linear.

3. Differential Operators

One-dimensional:

$$\frac{d}{dx}, \quad \frac{d^2}{dx^2}$$

Multivariable:

$$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}$$

General form:

$$\hat{L}u = \sum_{k=0}^{n} a_k \frac{d^k u}{dx^k}$$

4. Nabla Operator

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

Gradient:

$$\nabla u = (u_x, u_y, u_z)$$

Direction of the maximum increase of the function.
Divergence:

$$\nabla \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

Measures sources and sinks of a vector field.

5. Laplace Operator

$$\Delta = \nabla \cdot \nabla$$

$$\Delta u = u_{xx} + u_{yy} + u_{zz}$$

Characterizes the curvature of a field.

6. Integral Operator

$$(\hat{K}u)(x) = \int_a^b K(x,s)\,u(s)\,ds$$

Here $K(x,s)$ is called the kernel of the operator.

7. Fundamental Equations

Laplace equation:

$$\Delta u = 0$$

Poisson equation:

$$\Delta u = f$$

Heat equation:

$$u_t = \Delta u$$

Wave equation:

$$u_{tt} = c^2 \Delta u$$

8. Key Points

- an operator acts on a function
- the result is a new function
- linearity is fundamental
- ∇ is a first-order operator
- Δ is a second-order operator

Lecture 2. Differential Operators. Commutator

1 Differential Operator

A differential operator is a linear operator acting on functions by means of derivatives.

Examples:

$$\frac{\partial}{\partial x}, \quad \frac{\partial}{\partial y}, \quad \frac{\partial^2}{\partial x^2}, \quad \frac{\partial^2}{\partial x \partial y}$$

Action of the operator on a function $f(x, y, z)$:

$$\frac{\partial}{\partial x} f = \frac{\partial f}{\partial x}$$

2 Linearity

$$\frac{\partial}{\partial x}(af + bg) = a\frac{\partial f}{\partial x} + b\frac{\partial g}{\partial x}$$

3 Nabla Operator

The nabla operator is a vector differential operator:

$$\vec{\nabla} = \vec{e}_x \frac{\partial}{\partial x} + \vec{e}_y \frac{\partial}{\partial y} + \vec{e}_z \frac{\partial}{\partial z}$$

3.1 Gradient

For a scalar function $f(x, y, z)$:

$$\vec{\nabla} f = \vec{e}_x \frac{\partial f}{\partial x} + \vec{e}_y \frac{\partial f}{\partial y} + \vec{e}_z \frac{\partial f}{\partial z}$$

3.2 Divergence

Let the vector field be:

$$\vec{A} = \vec{e}_x A_x + \vec{e}_y A_y + \vec{e}_z A_z$$

Then:

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

3.3 Curl

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \vec{e}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \vec{e}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \vec{e}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

4 Laplacian

The Laplacian is a scalar second-order differential operator:

$$\Delta = \vec{\nabla} \cdot \vec{\nabla}$$

In Cartesian coordinates:

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

5 Sequential Action of Operators

$$\hat{A}\hat{B}f = \hat{A}(\hat{B}f)$$

Example:

$$\frac{\partial}{\partial x}\frac{\partial}{\partial y}f = \frac{\partial^2 f}{\partial x \partial y}$$

6 Commutator of Operators

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

For sufficiently smooth functions:

$$\left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right] = 0$$

Since:

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

6.1 Example of a Non-Zero Commutator

Let:

$$\hat{A} = x, \quad \hat{B} = \frac{\partial}{\partial x}$$

Then:

$$[x, \frac{\partial}{\partial x}]f = x\frac{\partial f}{\partial x} - \frac{\partial}{\partial x}(xf) = -f$$

Therefore:

$$[x, \frac{\partial}{\partial x}] = -1$$

Lecture 3. Eigenvalues and Eigenfunctions (Eigenvectors)

1 Why This Is Important in Mathematical Physics

Most problems of mathematical physics are reduced to solving a spectral problem of the form

$$\hat{L}f = \lambda f, \tag{1}$$

where $\hat{L}$ is a linear operator, f is a function, and λ is an eigenvalue.

Eigenfunctions describe the natural modes of a system, while eigenvalues determine the characteristic parameters of these modes (energy, frequency, decay rate, etc.).

2 Definition

A nonzero function $f \neq 0$ is called an *eigenfunction* of a linear operator $\hat{L}$ if

$$\hat{L}f = \lambda f. \tag{2}$$

The number λ is called the *eigenvalue* of the operator $\hat{L}$.

3 Finite-Dimensional Case

For a matrix A, eigenvalues and eigenvectors are defined by the equation

$$A\vec{x} = \lambda\vec{x}, \tag{3}$$

or equivalently,

$$(A - \lambda I)\vec{x} = 0. \tag{4}$$

A nontrivial solution exists if the characteristic equation

$$\det(A - \lambda I) = 0 \tag{5}$$

is satisfied.

4 Infinite-Dimensional Case

In problems of mathematical physics, operators act on functions and are usually differential operators:

$$\hat{L} = \frac{d^2}{dx^2}, \quad \hat{L} = -\Delta, \quad \hat{L} = -\frac{d^2}{dx^2} + V(x). \tag{6}$$

The spectral problem has the form

$$\hat{L}f(x) = \lambda f(x), \quad x \in \Omega, \tag{7}$$

and must be supplemented by boundary conditions.

5 Sturm–Liouville Problem

Example (a special case of the Sturm–Liouville problem)

Consider the problem

$$-\frac{d^2y}{dx^2} = \lambda y, \quad 0 < x < L, \tag{8}$$

with Dirichlet boundary conditions

$$y(0) = 0, \quad y(L) = 0. \tag{9}$$

The eigenvalues are

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \ldots \tag{10}$$

The corresponding eigenfunctions are

$$y_n(x) = \sin\frac{n\pi x}{L}. \tag{11}$$

6 Orthogonality of Eigenfunctions

Eigenfunctions corresponding to different eigenvalues are orthogonal:

$$\int_0^L y_n(x)\, y_m(x)\, dx = 0, \quad n \neq m. \tag{12}$$

For the considered example,

$$\int_0^L \sin\frac{n\pi x}{L} \sin\frac{m\pi x}{L}\, dx = \begin{cases} 0, & n \neq m, \\ \dfrac{L}{2}, & n = m. \end{cases} \tag{13}$$

7 Normalization of Eigenfunctions

Normalized eigenfunctions $\phi_n(x)$ are defined by the condition

$$\int_0^L |\phi_n(x)|^2\, dx = 1. \tag{14}$$

The normalized eigenfunctions are related to the original eigenfunctions by

$$\phi_n(x) = A_n\, y_n(x), \tag{15}$$

where A_n is a normalization constant (amplitude) chosen to satisfy the normalization condition.

Lecture 4. Expansion in Eigenfunctions. Fourier Series

1 Why Fourier Series Are Needed in Mathematical Physics

In many problems of mathematical physics it is necessary to represent an arbitrary function as an expansion in a basis of eigenfunctions of a linear operator. A classical example of such an expansion is given by trigonometric Fourier series.

Fourier series naturally arise in the solution of the heat equation, the wave equation, the Schrödinger equation, and other fundamental equations of physics.

2 General Trigonometric Fourier Series

2.1 Fourier Series on the Interval $[-\pi, \pi]$

Let $f(t)$ be a 2π-periodic function. Then its trigonometric Fourier series has the form

$$f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos nt + b_n \sin nt\right), \quad t \in [-\pi, \pi]. \tag{1}$$

The Fourier coefficients are defined by

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t)\, dt, \tag{2}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt)\, dt, \quad n \geq 1, \tag{3}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt)\, dt, \quad n \geq 1. \tag{4}$$

If the function $f(t)$ is piecewise smooth, then the Fourier series converges to $f(t)$ at all points of continuity.

3 Fourier Series with Period $2L$

Consider a function $f(x)$ that is periodic with period $2L$. Introduce the change of variables

$$t = \frac{\pi}{L} x.$$

Then the Fourier series can be written as

$$f(x) \sim \frac{A_0}{2} + \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi x}{L} + B_n \sin \frac{n\pi x}{L}\right), \quad x \in [-L, L]. \tag{5}$$

The coefficients are given by

$$A_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx, \tag{6}$$

$$A_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx, \tag{7}$$

$$B_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\frac{n\pi x}{L}\,dx. \tag{8}$$

4 Expansion of Functions on the Interval $[0, L]$

In practice, a function is often defined only on the semi-interval $[0, L]$. To apply Fourier series, it is extended to the symmetric interval $[-L, L]$.

4.1 Cosine Series (Even Extension)

Define the extension by

$$\tilde{f}(-x) = \tilde{f}(x).$$

In this case the function is expanded into a cosine series:

$$f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{L}, \quad x \in [0, L], \tag{9}$$

where

$$a_0 = \frac{2}{L}\int_0^L f(x)\,dx, \qquad a_n = \frac{2}{L}\int_0^L f(x)\cos\frac{n\pi x}{L}\,dx. \tag{10}$$

4.2 Sine Series (Odd Extension)

Define the extension by

$$\tilde{f}(-x) = -\tilde{f}(x).$$

In this case we obtain a sine series:

$$f(x) \sim \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}, \quad x \in [0, L], \tag{11}$$

where

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\frac{n\pi x}{L}\,dx. \tag{12}$$

5 Connection with Eigenfunctions of an Operator

Consider the operator

$$\hat{L} = -\frac{d^2}{dx^2}$$

defined on the interval $[0, L]$.

5.1 Dirichlet Boundary Conditions

Under the conditions

$$\phi(0) = \phi(L) = 0$$

the eigenfunctions have the form

$$\phi_n(x) = \sin \frac{n\pi x}{L}, \qquad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

5.2 Neumann Boundary Conditions

Under the conditions

$$\phi'(0) = \phi'(L) = 0$$

the eigenfunctions have the form

$$\phi_n(x) = \cos \frac{n\pi x}{L}, \qquad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

Thus, Fourier series represent expansions in the eigenfunctions of the second-derivative operator.

6 Orthogonality of Basis Functions

For sine functions:

$$\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L}\, dx = \frac{L}{2}\delta_{nm}. \tag{13}$$

For cosine functions:

$$\int_0^L \cos \frac{n\pi x}{L} \cos \frac{m\pi x}{L}\, dx = \frac{L}{2}\delta_{nm}. \tag{14}$$

7 Typical Example

Expand the function

$$f(x) = x, \quad x \in [0, L],$$

into a sine Fourier series.

The coefficients are

$$b_n = \frac{2}{L}\int_0^L x \sin \frac{n\pi x}{L}\, dx = \frac{2L}{n\pi}(-1)^{n+1}. \tag{15}$$

The final expansion is

$$x = \sum_{n=1}^{\infty} \frac{2L}{n\pi}(-1)^{n+1} \sin \frac{n\pi x}{L}. \tag{16}$$

8 Conclusion

Fourier series are a direct consequence of the spectral theory of linear operators. The form of the expansion is determined by the boundary conditions and the choice of the eigenfunction basis.

Lecture 5
Classification of Linear Second-Order Partial Differential Equations

1 General Form of the Equation

Consider a general linear second-order partial differential equation in two variables:

$$A(x,y)u_{xx} + B(x,y)u_{xy} + C(x,y)u_{yy} + Du_x + Eu_y + Fu = G. \tag{1}$$

The classification depends only on the highest-order terms:

$$Au_{xx} + Bu_{xy} + Cu_{yy}.$$

Define the discriminant:

$$\Delta = B^2 - 4AC. \tag{2}$$

If the coefficients depend on x, y, the classification is performed at each point of the domain.

2 Elliptic Equations ($\Delta < 0$)

Example:

$$u_{xx} + u_{yy} = 0. \tag{3}$$

This is the Laplace equation.

Properties

- describe stationary processes;
- no distinguished time direction;
- solutions are determined by boundary conditions;
- smoothing behavior of solutions.

Typical formulation: boundary value problem.

3 Parabolic Equations ($\Delta = 0$)

Example:

$$u_t = a^2 u_{xx}. \tag{4}$$

This is the heat equation.

Properties

- time evolution;
- smoothing of initial data;
- higher-frequency modes decay faster.

Typical formulation: initial-boundary value problem.

4 Hyperbolic Equations ($\Delta > 0$)

Example:

$$u_{tt} = c^2 u_{xx}. \tag{5}$$

This is the wave equation.

Properties

- propagation of disturbances;
- finite speed of signal transmission;
- oscillatory behavior of solutions;
- existence of characteristic curves.

Typical formulation: Cauchy problem.

5 Geometric Interpretation

Consider the quadratic form:

$$Adx^2 + Bdxdy + Cdy^2. \tag{6}$$

The type of the PDE coincides with the type of this quadratic form:

Sign of Δ	Type of Conic	Type of PDE
$\Delta < 0$	Ellipse	Elliptic
$\Delta = 0$	Parabola	Parabolic
$\Delta > 0$	Hyperbola	Hyperbolic

Lecture 6
Method of Separation of Variables. Boundary Value Problems

1 Problem Formulation

We consider partial differential equations on the interval

$$0 < x < L.$$

1. Heat Equation

$$u_t = a^2 u_{xx}.$$

2. Wave Equation

$$u_{tt} = c^2 u_{xx}.$$

Boundary conditions (Dirichlet):

$$u(0,t) = 0, \qquad u(L,t) = 0.$$

Initial conditions:
For the heat equation:

$$u(x,0) = f(x).$$

For the wave equation:

$$u(x,0) = f(x), \qquad u_t(x,0) = g(x).$$

2 Method of Separation of Variables

We seek a solution in the form

$$u(x,t) = X(x)T(t).$$

3 Heat Equation

Substituting:

$$XT' = a^2 X'' T.$$

Dividing variables:

$$\frac{T'}{a^2 T} = \frac{X''}{X}.$$

Since the left-hand side depends only on t and the right-hand side only on x, both must be equal to a constant:

$$\frac{T'}{a^2 T} = \frac{X''}{X} = -\lambda.$$

We obtain the system:

$$X'' + \lambda X = 0,$$

$$T' + a^2 \lambda T = 0.$$

3.1 Spatial Eigenvalue Problem

With boundary conditions:

$$X(0) = 0, \qquad X(L) = 0,$$

non-trivial solutions exist only for

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \ldots$$

Eigenfunctions:

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right).$$

3.2 Time-Dependent Part

$$T' + a^2 \lambda T = 0.$$

Separating variables:

$$\frac{dT}{T} = -a^2 \lambda dt.$$

Integrating:

$$\ln|T| = -a^2 \lambda t + C.$$

$$T(t) = Ce^{-a^2 \lambda t}.$$

Final solution:

$$u(x,t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi x}{L}\right) e^{-a^2 \left(\frac{n\pi}{L}\right)^2 t}.$$

Coefficients:

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

4 Wave Equation

Substituting:

$$XT'' = c^2 X'' T.$$

Separating variables:

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -\lambda.$$

We obtain:

$$X'' + \lambda X = 0,$$
$$T'' + c^2 \lambda T = 0.$$

4.1 Time-Dependent Part

Define

$$\omega_n = \frac{n\pi c}{L}.$$

Then:

$$T'' + \omega_n^2 T = 0.$$

Solution:

$$T_n(t) = A_n \cos(\omega_n t) + B_n \sin(\omega_n t).$$

General solution:

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[A_n \cos(\omega_n t) + B_n \sin(\omega_n t)\right].$$

4.2 Determination of Coefficients

From $u(x,0) = f(x)$:

$$A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

Compute time derivative:

$$u_t(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[-A_n \omega_n \sin(\omega_n t) + B_n \omega_n \cos(\omega_n t)\right].$$

At $t = 0$:

$$u_t(x,0) = \sum_{n=1}^{\infty} B_n \omega_n \sin\left(\frac{n\pi x}{L}\right).$$

Thus,

$$B_n = \frac{2}{\omega_n L} \int_0^L g(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

5 Physical Interpretation

For the heat equation: each mode decays exponentially in time.

For the wave equation: standing waves arise with natural frequencies

$$\omega_n = \frac{n\pi c}{L}.$$

Lecture 7: Laplace Equation. Dirichlet Problem

1 Laplace Equation

One of the fundamental equations of mathematical physics is the **Laplace equation**

$$\Delta u = 0$$

where

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

is the Laplace operator.
In two dimensions the equation takes the form

$$u_{xx} + u_{yy} = 0$$

Functions satisfying this equation are called **harmonic functions**.

2 Physical Meaning

The Laplace equation appears in many problems of physics.
Electrostatics

$$\Delta \varphi = 0$$

when there are no electric charges in the region.
Steady-state heat conduction

$$\Delta T = 0$$

when the temperature distribution is stationary and there are no heat sources.
Thus the Laplace equation describes **equilibrium fields without sources**.

3 Boundary Value Problems

To determine the solution uniquely, boundary conditions must be specified.

3.1 Dirichlet Problem

The value of the function is given on the boundary of the domain:

$$u|_{\partial\Omega} = f$$

For example, this may represent a prescribed temperature on the boundary of a body.

4 Problem Formulation

Consider a rectangular region

$$0 < x < L, \qquad 0 < y < a$$

and the Laplace equation

$$u_{xx} + u_{yy} = 0$$

with boundary conditions

$$u(0, y) = 0$$

$$u(L, y) = 0$$

$$u(x, 0) = 0$$

$$u(x, a) = f(x)$$

This is the **Dirichlet problem** for the Laplace equation.

5 Method of Separation of Variables

We seek the solution in the form

$$u(x, y) = X(x)Y(y)$$

Substituting into the equation gives

$$X''Y + XY'' = 0$$

Dividing by XY yields

$$\frac{X''}{X} + \frac{Y''}{Y} = 0$$

Hence

$$\frac{X''}{X} = -\frac{Y''}{Y} = -\lambda$$

which leads to the system

$$X'' + \lambda X = 0$$

$$Y'' - \lambda Y = 0$$

6 Eigenvalue Problem for $X(x)$

From the boundary conditions

$$X(0) = 0, \qquad X(L) = 0$$

we obtain the problem

$$X'' + \lambda X = 0$$

Nontrivial solutions exist only for

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2$$

The corresponding eigenfunctions are

$$X_n(x) = \sin\frac{n\pi x}{L}$$

7 Solution for $Y(y)$

Substituting λ_n gives

$$Y'' - \left(\frac{n\pi}{L}\right)^2 Y = 0$$

The general solution is

$$Y_n(y) = A_n \cosh\frac{n\pi y}{L} + B_n \sinh\frac{n\pi y}{L}$$

8 Applying Boundary Conditions

From

$$u(x, 0) = 0$$

it follows that

$$A_n = 0$$

thus

$$Y_n(y) = B_n \sinh\frac{n\pi y}{L}$$

9 General Solution

The solution can therefore be written as

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sinh\frac{n\pi y}{L} \sin\frac{n\pi x}{L}$$

10 Determination of Coefficients

Using the boundary condition

$$u(x, a) = f(x)$$

we obtain

$$f(x) = \sum_{n=1}^{\infty} B_n \sinh \frac{n\pi a}{L} \sin \frac{n\pi x}{L}$$

The function $f(x)$ can be expanded into a Fourier sine series

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} \, dx$$

Thus

$$B_n = \frac{b_n}{\sinh \frac{n\pi a}{L}}$$

11 Final Solution

The solution of the Dirichlet problem is

$$u(x, y) = \sum_{n=1}^{\infty} \frac{b_n}{\sinh \frac{n\pi a}{L}} \sinh \frac{n\pi y}{L} \sin \frac{n\pi x}{L}$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} \, dx$$

Lecture 8
Laplace Equation in Polar Coordinates

1. Polar Coordinates

$$x = \rho\cos\varphi, \quad y = \rho\sin\varphi$$

1.1 Derivation via Lamé Coefficients

The Laplacian in orthogonal curvilinear coordinates is:

$$\Delta u = \frac{1}{h_1 h_2}\left[\frac{\partial}{\partial q_1}\left(\frac{h_2}{h_1}\frac{\partial u}{\partial q_1}\right) + \frac{\partial}{\partial q_2}\left(\frac{h_1}{h_2}\frac{\partial u}{\partial q_2}\right)\right]$$

For polar coordinates:

$$q_1 = \rho, \quad q_2 = \varphi$$

$$h_\rho = 1, \quad h_\varphi = \rho$$

Hence:

$$\Delta u = \frac{1}{\rho}\left[\frac{\partial}{\partial \rho}\left(\rho\frac{\partial u}{\partial \rho}\right) + \frac{\partial}{\partial \varphi}\left(\frac{1}{\rho}\frac{\partial u}{\partial \varphi}\right)\right]$$

$$\Delta u = u_{\rho\rho} + \frac{1}{\rho}u_\rho + \frac{1}{\rho^2}u_{\varphi\varphi}$$

2. Laplace Equation

$$\Delta u = 0$$

$$u_{\rho\rho} + \frac{1}{\rho}u_\rho + \frac{1}{\rho^2}u_{\varphi\varphi} = 0$$

3. Separation of Variables

$$u(\rho,\varphi) = R(\rho)\Phi(\varphi)$$

$$R''\Phi + \frac{1}{\rho}R'\Phi + \frac{1}{\rho^2}R\Phi'' = 0$$

Divide:

$$\frac{R''}{R} + \frac{1}{\rho}\frac{R'}{R} + \frac{1}{\rho^2}\frac{\Phi''}{\Phi} = 0$$

Multiply by ρ^2:

$$\rho^2 \frac{R''}{R} + \rho \frac{R'}{R} + \frac{\Phi''}{\Phi} = 0$$

4. Separation

$$\frac{\Phi''}{\Phi} = -\lambda$$

$$\rho^2 R'' + \rho R' - \lambda R = 0$$

5. Angular Equation

$$\Phi'' + \lambda \Phi = 0$$

Periodic condition:

$$\Phi(\varphi + 2\pi) = \Phi(\varphi) \Rightarrow \lambda = n^2, \quad n \in \mathbb{Z}$$

$$\Phi(\varphi) = A \cos n\varphi + B \sin n\varphi$$

6. Radial Equation

$$\rho^2 R'' + \rho R' - n^2 R = 0$$

$$R(\rho) = C\rho^n + D\rho^{-n}$$

7. General Solution

$$u(\rho, \varphi) = \sum_{n=0}^{\infty} \left(C_n \rho^n + D_n \rho^{-n}\right) \left(A_n \cos n\varphi + B_n \sin n\varphi\right)$$

8. Solution in a Disk

For $\rho \leq R$, boundedness at $\rho = 0$:

$$D_n = 0$$

$$u(\rho, \varphi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \rho^n \left(A_n \cos n\varphi + B_n \sin n\varphi\right)$$

9. Boundary Condition

$$u(R, \varphi) = f(\varphi)$$

$$u(\rho, \varphi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{\rho}{R}\right)^n \left(a_n \cos n\varphi + b_n \sin n\varphi\right)$$

10. Fourier Coefficients

$$f(\varphi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\varphi + b_n \sin n\varphi)$$

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(\varphi)\, d\varphi$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \cos n\varphi\, d\varphi$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(\varphi) \sin n\varphi\, d\varphi$$

11. Final Solution

$$u(\rho, \varphi) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(\frac{\rho}{R}\right)^n (a_n \cos n\varphi + b_n \sin n\varphi)$$

12. Examples

Example 1:

$$u(R, \varphi) = \cos 2\varphi$$

$$u(\rho, \varphi) = \left(\frac{\rho}{R}\right)^2 \cos 2\varphi$$

Example 2:

$$u(R, \varphi) = 3 + 2\cos\varphi - \sin 3\varphi$$

$$u(\rho, \varphi) = 3 + 2\left(\frac{\rho}{R}\right) \cos\varphi - \left(\frac{\rho}{R}\right)^3 \sin 3\varphi$$

13. Key Idea

Each boundary harmonic:

$$\cos n\varphi, \quad \sin n\varphi$$

extends into the disk as:

$$\left(\frac{\rho}{R}\right)^n \cos n\varphi, \quad \left(\frac{\rho}{R}\right)^n \sin n\varphi$$

Higher modes decay faster toward the center.

Lecture 9
Laplace Equation in Spherical Coordinates and its Connection with Angular Momentum Operator

1. Introduction

We consider the Laplace equation:

$$\Delta u = 0$$

It arises in electrostatics, gravitation, and steady-state heat conduction.

2. Spherical Coordinates

$$x = r\sin\theta\cos\varphi, \quad y = r\sin\theta\sin\varphi, \quad z = r\cos\theta$$

3. Laplacian in Spherical Coordinates

$$\Delta = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

4. Angular Momentum Operator

In quantum mechanics, we introduce the operator:

$$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$$

Thus, the Laplacian can be written as:

$$\Delta = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{L^2}{\hbar^2 r^2}$$

5. Separation of Variables

We seek a solution in the form:

$$u(r,\theta,\varphi) = R(r)Y(\theta,\varphi)$$

Substituting into the Laplace equation:

$$\frac{1}{r^2}Y\frac{d}{dr}(r^2R') - \frac{R}{\hbar^2 r^2}L^2Y = 0$$

Dividing by RY:

$$\frac{1}{R}\frac{d}{dr}(r^2R') - \frac{1}{\hbar^2}\frac{L^2Y}{Y} = 0$$

6. Spectral Problem

The angular part gives:

$$L^2Y = \hbar^2 l(l+1)Y$$

where $l = 0, 1, 2, \ldots$

7. Radial Equation

$$\frac{d}{dr}(r^2 R') - l(l+1)R = 0$$

8. Solution of the Radial Equation

$$R(r) = Ar^l + \frac{B}{r^{l+1}}$$

9. Angular Functions

The solutions are spherical harmonics:

$$Y_l^m(\theta, \varphi)$$

where $m = -l, \ldots, l$

10. General Solution

$$u(r, \theta, \varphi) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \left(A_{lm} r^l + \frac{B_{lm}}{r^{l+1}} \right) Y_l^m(\theta, \varphi)$$

11. Behavior of Solutions

As $r \to 0$:

$$u \sim r^l$$

As $r \to \infty$:

$$u \sim \frac{1}{r^{l+1}}$$

12. Physical Meaning

Spherical harmonics are eigenfunctions of the operator L^2:

$$L^2 Y_l^m = \hbar^2 l(l+1) Y_l^m$$

The solution of the Laplace equation is expressed as an expansion in these functions.

13. Simple Cases

- $l = 0$: $u \sim \frac{1}{r}$ (monopole)
- $l = 1$: dipole
- $l = 2$: quadrupole

Conclusion

The Laplace equation in spherical coordinates reduces to a spectral problem for the angular momentum operator. The solution is constructed using the eigenfunctions of L^2.

Lecture 10
Dirac Delta Function

1. Introduction

In mathematical physics, it is often necessary to describe a source concentrated at a single point. For this purpose, the Dirac delta function is introduced.

It is not an ordinary function, but is defined through its properties under the integral.

2. Formal definition

$$\delta(x - x_0) = \begin{cases} \infty, & x = x_0 \\ 0, & x \neq x_0 \end{cases}$$

and

$$\int_{-\infty}^{\infty} \delta(x - x_0)\, dx = 1$$

3. Fundamental property

$$\int_{-\infty}^{\infty} \delta(x - x_0)\, \varphi(x)\, dx = \varphi(x_0)$$

Thus, the delta function "selects" the value of a function at the point x_0.

4. Properties

The main properties of the delta function are:

1) Normalization

$$\int \delta(x - x_0)\, dx = 1$$

2) Localization property

$$\delta(x - x_0) = 0, \quad x \neq x_0$$

3) Sampling property

$$\int \delta(x - x_0)\, \varphi(x)\, dx = \varphi(x_0)$$

4) Multiplication by a function

$$f(x)\delta(x - x_0) = f(x_0)\delta(x - x_0)$$

5) Scaling property

$$\delta(a(x - x_0)) = \frac{1}{|a|}\delta(x - x_0)$$

5. Heaviside function

$$H(x - x_0) = \begin{cases} 0, & x < x_0 \\ 1, & x > x_0 \end{cases}$$

$$\frac{d}{dx} H(x - x_0) = \delta(x - x_0)$$

6. Derivative of the delta function

$$\int_{-\infty}^{\infty} \delta'(x - x_0)\, \varphi(x)\, dx = -\varphi'(x_0)$$

7. Delta function of a function

Let $f(x)$ be a smooth function with simple zeros x_i, such that

$$f(x_i) = 0.$$

Then the delta function can be represented as

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|}.$$

Interpretation.

The delta function is nonzero only at the points where $f(x) = 0$. If there are several such points, it splits into a sum of delta functions, each localized at a root x_i.

The factor $1/|f'(x_i)|$ accounts for the local stretching of the variable.

Example:

$$f(x) = x^2 - 1$$

The zeros are $x = \pm 1$, and $f'(x) = 2x$. Therefore,

$$\delta(x^2 - 1) = \frac{1}{2}\delta(x - 1) + \frac{1}{2}\delta(x + 1).$$

8. Three-dimensional delta function

$$\iiint \delta(\mathbf{r} - \mathbf{r}_0)\, \varphi(\mathbf{r})\, dV = \varphi(\mathbf{r}_0)$$

9. Cartesian coordinates

$$\delta(\mathbf{r} - \mathbf{r}_0) = \delta(x - x_0)\, \delta(y - y_0)\, \delta(z - z_0)$$

10. Normalization

$$\iiint \delta(\mathbf{r} - \mathbf{r}_0)\, dV = 1$$

11. Spherical coordinates

$$dV = r^2 \sin\theta\, dr\, d\theta\, d\varphi$$

General form:

$$\delta(\mathbf{r} - \mathbf{r}_0) = \frac{1}{r^2 \sin\theta} \delta(r - r_0)\, \delta(\theta - \theta_0)\, \delta(\varphi - \varphi_0)$$

Non-shifted case:

$$\delta(\mathbf{r}) = \frac{\delta(r)}{4\pi r^2}$$

12. Poisson equation

$$\Delta\varphi(\mathbf{r}) = -\frac{q}{\varepsilon_0}\delta(\mathbf{r}-\mathbf{r}_0)$$

$$\Delta\frac{1}{|\mathbf{r}-\mathbf{r}_0|} = -4\pi\delta(\mathbf{r}-\mathbf{r}_0)$$

Comparing these expressions, we obtain the solution:

$$\varphi(\mathbf{r}) = \frac{q}{4\pi\varepsilon_0}\frac{1}{|\mathbf{r}-\mathbf{r}_0|}$$

The obtained function is the potential of a point charge. It depends only on the distance from the source and decreases as $1/r$.

13. Conclusion

- the delta function describes a localized source,
- it is defined through its integral property.

Lecture 11-12
Spherical Waves

1. Wave Equation

Consider the wave equation in three-dimensional space:

$$u_{tt} = c^2 \Delta u.$$

In spherical coordinates, the Laplace operator is:

$$\Delta u = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial u}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2 u}{\partial \varphi^2}.$$

2. Spherical Symmetry

Assume the solution depends only on the radial coordinate:

$$u = u(r, t).$$

Then the equation simplifies to:

$$u_{tt} = c^2 \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial u}{\partial r}\right).$$

Expanding the derivative:

$$u_{tt} = c^2 \left(u_{rr} + \frac{2}{r} u_r\right).$$

3. Substitution

Introduce the substitution:

$$u(r, t) = \frac{1}{r} w(r, t).$$

Then:

$$u_r = -\frac{1}{r^2} w + \frac{1}{r} w_r,$$

$$u_{rr} = \frac{2}{r^3} w - \frac{2}{r^2} w_r + \frac{1}{r} w_{rr}.$$

Hence:

$$u_{rr} + \frac{2}{r} u_r = \frac{1}{r} w_{rr}.$$

4. One-Dimensional Equation

We obtain:

$$\frac{1}{r} w_{tt} = c^2 \frac{1}{r} w_{rr}.$$

Therefore:

$$w_{tt} = c^2 w_{rr}.$$

5. General Solution

The general solution is:

$$w(r,t) = f(r-ct) + g(r+ct).$$

Hence:

$$u(r,t) = \frac{1}{r}\big[f(r-ct) + g(r+ct)\big].$$

6. Incoming and Outgoing Waves

Outgoing wave:

$$u_{\text{out}}(r,t) = \frac{1}{r} f(r-ct).$$

Incoming wave:

$$u_{\text{in}}(r,t) = \frac{1}{r} g(r+ct).$$

7. Harmonic Solutions

Assume a solution of the form:

$$w(r,t) = e^{i(kr-\omega t)}.$$

Substitution gives:

$$\omega^2 = c^2 k^2.$$

Then:

$$u_{\text{out}} = \frac{A}{r} e^{i(kr-\omega t)},$$

$$u_{\text{in}} = \frac{B}{r} e^{i(-kr-\omega t)}.$$

8. Real Solutions

$$u_{\text{out}} = \frac{A}{r}\cos(kr-\omega t),$$

$$u_{\text{in}} = \frac{B}{r}\cos(kr+\omega t).$$

9. Physical Meaning

The amplitude decays as:

$$u \sim \frac{1}{r}.$$

This is because the energy is distributed over the surface of a sphere:

$$S = 4\pi r^2.$$

10. Summary

The general solution for a spherical wave is:

$$u(r,t) = \frac{1}{r}\big[f(r-ct) + g(r+ct)\big].$$

A spherical wave can be viewed as a one-dimensional wave multiplied by the factor $1/r$, which accounts for the geometry of space.

Lecture 13
Poisson Equation and Green's Function

1. Poisson Equation

Consider the equation

$$\Delta u(\mathbf{r}) = f(\mathbf{r}),$$

where Δ is the Laplace operator and $f(\mathbf{r})$ is a given function.

Special case:

$$f = 0 \;\Rightarrow\; \Delta u = 0,$$

which gives the Laplace equation.

2. Physical Meaning

The Poisson equation describes a field generated by a source (charge, mass, etc.).

3. Representation via the δ-function

Any function can be written as:

$$f(\mathbf{r}) = \int f(\mathbf{r}')\,\delta(\mathbf{r}-\mathbf{r}')\,d^3r'.$$

4. Green's Function

Define the Green's function:

$$\Delta G(\mathbf{r},\mathbf{r}') = \delta(\mathbf{r}-\mathbf{r}').$$

5. Solution via Green's Function

We seek a solution in the form:

$$u(\mathbf{r}) = \int G(\mathbf{r},\mathbf{r}')f(\mathbf{r}')\,d^3r'.$$

Verification:

$$\Delta u = f(\mathbf{r}).$$

6. Radial Symmetry

In an infinite space:

$$G(\mathbf{r},\mathbf{r}') = G(R), \quad R = |\mathbf{r}-\mathbf{r}'|.$$

7. Equation for G

For $R \neq 0$:

$$\Delta G = 0.$$

In spherical coordinates:

$$\frac{1}{R^2}\frac{d}{dR}\left(R^2\frac{dG}{dR}\right) = 0.$$

8. Solution

$$R^2 \frac{dG}{dR} = C_1, \quad \frac{dG}{dR} = \frac{C_1}{R^2}.$$

$$G(R) = -\frac{C_1}{R} + C_2.$$

Taking $C_2 = 0$:

$$G(R) = \frac{C}{R}.$$

9. Fundamental Identity

$$\Delta \frac{1}{r} = -4\pi\delta(\mathbf{r}).$$

Hence:

$$\Delta \left(-\frac{1}{4\pi r} \right) = \delta(\mathbf{r}).$$

10. Green's Function

$$\boxed{G(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}}$$

11. Solution of the Poisson Equation

$$\boxed{u(\mathbf{r}) = -\frac{1}{4\pi} \int \frac{f(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, d^3r'}$$

12. Summary

- Poisson equation:

$$\Delta u(\mathbf{r}) = f(\mathbf{r})$$

- Green's function:

$$\Delta G(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}')$$

- In three-dimensional space:

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|}$$

  In particular, for $\mathbf{r}' = 0$:

$$G(\mathbf{r}) = -\frac{1}{4\pi r}$$

Lecture 14–15
Summary of the Course in Mathematical Physics

1. General Idea of the Course

Mathematical physics follows the scheme:

$$\text{Physical problem} \;\Rightarrow\; \text{differential equation} \;\Rightarrow\; \text{expansion in a basis}$$

2. Operators

Linear operator:

$$\hat{L}(au + bv) = a\hat{L}u + b\hat{L}v$$

Examples:

$$\frac{d}{dx}, \quad \nabla, \quad \Delta$$

Commutator:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

3. Eigenvalues and Eigenfunctions

Main equation:

$$\hat{L}\psi = \lambda\psi$$

Physical meaning:

- oscillation frequencies
- energy levels
- system modes

4. Expansion in Eigenfunctions

$$f(x) = \sum_n c_n \psi_n(x)$$

Coefficients:

$$c_n = \int f(x)\psi_n(x)\,dx$$

5. Classification of Equations

- Elliptic:

$$\Delta u = 0$$

- Parabolic:

$$u_t = \Delta u$$

- Hyperbolic:

$$u_{tt} = \Delta u$$

6. Separation of Variables

$$u(x,t) = X(x)T(t)$$

Leads to:

$$\frac{T'}{T} = \frac{X''}{X} = -\lambda$$

7. Main Equations

Heat equation:

$$u_t = \alpha \Delta u$$

Wave equation:

$$u_{tt} = c^2 \Delta u$$

Laplace equation:

$$\Delta u = 0$$

Poisson equation:

$$\Delta u = f$$

8. Coordinate Systems and Bases

- Cartesian: $\sin, \cos$
- Polar: Bessel functions
- Spherical: $Y_l^m(\theta, \phi)$

9. Delta Function

$$\delta(x)$$

Property:

$$\int f(x)\delta(x - x_0)dx = f(x_0)$$

10. Green's Function

$$\Delta G(\mathbf{r}) = \delta(\mathbf{r})$$

Solution:

$$u(\mathbf{r}) = \int G(\mathbf{r}, \mathbf{r}')f(\mathbf{r}')d^3r'$$

11. Spherical Waves

$$u(r,t) = \frac{1}{r}f(r \pm ct)$$

12. Main Structure

$$\text{Operator} \Rightarrow \text{eigenfunctions} \Rightarrow \text{expansion} \Rightarrow \text{solution}$$

13. Physical Meaning

Any system can be represented as a sum of independent modes.

14. Connection to Other Fields

- quantum mechanics
- electrodynamics
- field theory
- classical mechanics

15. Final Statement

$$\boxed{\text{World} = \text{operators} + \text{spectrum} + \text{symmetry}}$$